

Serbia Team Selection Test Round 1 2005 — P3

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Find all polynomials $P(x)$ that satisfy $P(x^2 + 1) = P(x)^2 + 1$ for all x .

Solution

Lemma: Any polynomial $P(x) \neq x$ has finitely many fixed points.

Proof: Suppose not, then $P(x) - x$ has infinitely many solutions, but since the degree of the polynomial $P(x) - x$ is finite this is a contradiction. $\square$

Clearly $P(x) = x$ is a solution, now let's find polynomials $P(x) \neq x$ that work also.

Claim: If t is a fixed point of the polynomial P , then so is $t^2 + 1$.

Proof: By definition of a fixed point and using the given equation gives $P(t^2 + 1) = P(t)^2 + 1 = t^2 + 1$ and so $t^2 + 1$ is a fixed point. $\square$

Now suppose that some complex number z is a fixed point of P . We must have $|z| = |z^2 + 1|$ in order to avoid having infinitely many fixed points. If we write $z = a + bi$ for some real a, b then that last equation is equivalent to $a^2 + b^2 = (a^2 - b^2 + 1)^2 + (2ab)^2$. If we think of this equation as a polynomial of, say a , the polynomial is of degree 4 and there are at most 4 solutions. The 4 solutions are $z = \frac{1}{2} \pm \frac{\sqrt{3}i}{2}, -\frac{1}{2} \pm \frac{\sqrt{3}i}{2}$ and these are easily verifiable since they correspond to the roots of $x^2 - x + 1 = 0$ and $x^2 + x + 1 = 0$. Since these are the only fixed points of P we can write that $P(x) - x = (x^2 - x + 1)(x^2 + x + 1)$ or $P(x) - x = x^2 - x + 1$ or $P(x) - x = x^2 + x + 1$ since we know that roots come in complex conjugate pairs. Solve for $P(x)$ in each of these equations to show that the candidates for $P(x)$ are $x^4 + x^2 - x + 1$, $(x+1)^2$ and $x^2 + 1$. Then finally one can check that the only solutions are $P(x) = x$ or $x^2 + 1$.

Remark: Note that this solution is actually only partially correct. Instead of the condition $|z| = |z^2 + 1|$ the complete condition is actually that the sequence $z, z^2 + 1, (z^2 + 1)^2 + 1, \dots$ must have some finite period, and this means that there are finitely many fixed points as we initially desired. Once you work this logic out the actual answer is $P(x) = x$ or $x^2 + 1$ or $x^2 + 1$ composed with itself any positive integer number of times.